

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0108

Quantico 138 kV
April 20, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0108. The Interconnection Customer proposes a 114.8 MW solar photovoltaic facility interconnecting at the Quantico 138 kV substation in the Virginia area. The request was received on February 20, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 91 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$72,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Nasa - Bunker Hill 138kV L
Contingency: Taber - Dixleta 345kV
Loading: 106.0% (Rating: 238 MVA)
DFAX: 5.3%
Pre-project loading: 103.5%
2. Facility: Fairland Substation - Sunbright 138kV
Contingency: Adrian Energy Associates Llc 345-138 kV XFMR
Loading: 104.6% (Rating: 230 MVA)
DFAX: 6.0%
Pre-project loading: 101.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Nasa - Bunker Hill 138kV L	Reconductor line	\$883,000	7% (\$65,000)
Fairland Substation - Sunbright 138kV	Reconductor line	\$165,000	4% (\$7,000)
	Total allocated to project		\$72,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.